

Arshiya Dilkush

Data Engineer

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SUMMARY

Experienced Data Engineer with around 3+ years of experience, specializing in designing, building, and optimizing scalable data pipelines across cloud and big data ecosystems, supporting analytics, reporting, and downstream data consumers. Proven collaborator with strong ownership of ingestion, transformation, orchestration, and data quality processes, delivering reliable datasets through disciplined engineering practices and cross-functional teamwork.

SKILLS

Cloud & Data Platforms: Microsoft Azure (Azure Data Lake Storage, Azure Databricks, Azure Synapse Analytics, Azure Data Factory)

Big Data Processing: Apache Spark (PySpark), Distributed Batch Processing, Micro-Batch Processing, Performance Optimization

Data Ingestion & Streaming: Apache Kafka, Apache NiFi, Sqoop, SFTP-based Ingestion, Change Data Capture (CDC)

Data Storage & Warehousing: Delta Lake, Hadoop HDFS, Hive, Presto, PostgreSQL, Snowflake

Data Modeling & Transformation: Dimensional Modeling, Canonical Data Models, Schema Evolution, Data Normalization, Incremental Loads

Workflow Orchestration: Apache Airflow, DAG Design, Scheduling, SLA Monitoring, Failure Recovery

Data Quality & Monitoring: Data Validation Rules, Data Reconciliation, Pipeline Health Monitoring, Anomaly Detection, Logging & Alerts

Programming & Scripting: Python (PySpark), SQL (Advanced), Linux Shell Scripting

Security & Governance: Role-Based Access Control, Data Encryption at Rest and In Transit, Audit Readiness, Compliance-Aligned Data Handling

Visualization & Analytics Integration: Power BI, Presto SQL, Analytical Extracts for Reporting and Dashboards

EXPERIENCE

Eleven Health, USA | Data Engineer

Aug 2025– Present

- Architected unified healthcare lakehouse using Kafka, Databricks, and Delta Lake to integrate clinical and claims data, improving analytics and ML data availability by 45%.
- Engineered real-time pipelines with Spark Structured Streaming, Kafka, and AWS S3, applying HL7 normalization, ICD-10 mapping, and HIPAA controls, reducing data latency by 60%.
- Developed scalable transformation layers using Databricks, Spark, and dbt, enabling incremental modeling and schema evolution, increasing data reliability and consistency by 50%.
- Curated feature-ready datasets with Snowflake, dbt, and Delta Lake to support predictive healthcare models, improving high-risk patient identification accuracy by 35%.
- Implemented data quality and observability using Great Expectations, Monte Carlo, and Snowflake, reducing downstream clinical and revenue reporting issues by 70%.
- Automated secure cloud infrastructure with Terraform, AWS IAM, and Kubernetes, accelerating compliant environment provisioning by 40% while enforcing HIPAA-aligned access controls.

Mphasis, India | Data Engineer

May 2021– Nov 2023

- Architected an integrated healthcare data platform using Azure Data Lake, Apache Spark, and Kafka to consolidate clinical and billing sources, improving enterprise data availability by 45%.
- Engineered ingestion and transformation pipelines with Apache NiFi, Spark, and Hive, standardizing patient and billing records, reducing downstream data inconsistencies by 55% across hospital analytics workflows.
- Developed scalable batch processing and curated analytical layers using Spark, Hive, and Presto, enabling operational reporting for capacity and revenue insights with query performance improved by 40%.
- Improved billing reconciliation accuracy by 35% by designing matching logic using Spark, PostgreSQL, and SQL, enabling faster identification of payment gaps for hospital revenue teams.
- Reduced manual billing intervention by 75% by automating file parsing and normalization through Talend, HDFS, and Linux scripting, delivering cleaner reconciliation datasets to finance operations.
- Increased pipeline reliability by 60% by orchestrating end-to-end workflows with Apache Airflow, Azure monitoring, and structured alerting, ensuring timely data delivery for daily hospital reporting.
- Enhanced audit readiness and data governance by 50% by implementing validation checks, retention strategies, and secure access controls using Azure security, Spark validation frameworks, and healthcare data modeling practices.

EDUCATION

Avila University, USA | *Master of Science in Technology Management in Data Science*

Dec 2025

CERTIFICATES

Python Certification | Intermediate Machine Learning- Kaggle | Deep Learning Certification — Simplilearn